SHIVAM VIPUL KAPADIA

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EDUCATION

University of Illinois at Urbana-Champaign Expected May 2029
Bachelor of Science in Computer Engineering with a Data Science Minor GPA: 4.0

- Honors: James Scholar, Dean's List Fall 2025
- **ECE 120:** Digital Logic, Von-Nuemann Model, Finite State Machine; **ECE 110:** Circuit Analysis, Electronic Devices, Digital and Analog systems; **ECE 145:** Engineering Design, Hardware–Software integration, Prototyping; **ECE 220:** Systems Knowledge, Pointers, Arrays, Data Structures, Recursion; **PHY 211:** Mechanics; **MATH 241**: Multivariate Calculus, **MATH 285:** Differential Equations

Atul Vidyalaya May 2025
Indian Certificate for Secondary Education and Indian School Certificate High School Curriculum Final Grade: 98.2%

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, Tableau, HTML, CSS, JavaScript, LC-3 Assembly, C, C++
Frameworks/Tools: GitHub, Git, Xilinx Vivado, CAD
Spoken Languages: English (Full Working Proficiency), Hindi, Gujarati

WORK EXPERIENCE

1Rivet India
Software Development Intern May 2024-May 2024
• Utilized theoretical knowledge practically for creating a user interface of a manufacturing company and improving on its visual and interactive components using HTML and CSS
• Deployed algorithms for querying a database to accurately and efficiently store and retrieve the company's data by implementing SQL

ValU Wealth Care India
Data Analysis Intern Dec 2023–Feb 2024
• Utilized Tableau to create data visualizations to comprehensively understand financial data and trends of multiple penny stocks of Iron and Steel Industries including, Supreme Engineering Ltd., Hilton Metal Forging and SAL Steel
• Analyzed the data and determined their future growth potential by studying their performance over the years
• Assisted the firm with crucial data that helped in making strategic investment decisions, contributing to client satisfaction

PROJECT

Hands-Free Dorm-friendly Automatic Dish-washer Since January 2025
• Designing an Arduino-controlled, hands-free dishwashing system using a finite-state machine to coordinate sensing, clamping, scrubbing, soap dispensing, and rinsing.
• Building a compact, dorm-friendly appliance with servo actuation, rotating brush mechanisms, and relay-controlled water flow focused on reliability and safety.

BeatBot - Member at ASME Special Projects for EOH Since January 2025
• Developing a robotic drumming arm and stick mechanism capable of playing programmed song.
• Primarily working on the ECE team; simultaneously make contributions to CAD design.

School Library Management May 2024-June 2024
• Developed a highly efficient School Library Management System using Java and SQL, aimed at modernizing and automating all library operations. Currently the system has been deployed and is active, managing a diverse collection of 10,000 books, and ensuring seamless access for 800 students and faculty members

RESEARCH WORK

Comparative Analysis of Machine Learning Algorithms for air quality index prediction
Published in Machine Learning for Computational Science and Engineering, Springer (Feb 2025, Volume 1 Article 14) 10.1007/s44379-025-00014-2
• Studied 50+ different research papers on machine learning methods like KNN and SVM to better understand how we can improve AQI prediction models, and found that the hybrid models worked the best for predictions from review of multiple papers

Comparative Analysis of Machine Learning Algorithms like SVM and RF for rainfall prediction Currently Under Review
• Conducted extensive research on rainfall prediction based on machine learning algorithms like SVM and RF. Referred to 70 published papers focusing on Machine Learning and its application in weather forecasting